

Chapter 6

Lecture 6: Topological Spaces

6.1 Topological Spaces in general

Definition 6.1.1. A **topological space** is a pair $(X, \mathcal{U})$ where X is a set and $\mathcal{U} \subseteq 2^X$ such that:

1. $\emptyset, X \in \mathcal{U}$.
2. If $U_\alpha \in \mathcal{U}$ for all $\alpha \in A$, then $\bigcup_{\alpha \in A} U_\alpha \in \mathcal{U}$.
3. If $U_1, U_2, \dots, U_n \in \mathcal{U}$, then $\bigcap_{i=1}^n U_i \in \mathcal{U}$.

$\mathcal{U}$ is called a **topology** on X . The elements of $\mathcal{U}$ are called **open sets**.

Example 6.1.1. Any metric space (X, d) is a topological space, with $\mathcal{U}$ defined as:

$$U \in \mathcal{U} \iff \forall x \in U, \exists r > 0 \text{ such that } B(x, r) \subseteq U. \quad (6.1)$$

Open sets defined in this topology are the same as open sets defined by the metric.

Example 6.1.2. Let X be any non-empty set. The **discrete topology** on X is defined as $\mathcal{U} = 2^X$. Every subset of X is open.

Example 6.1.3. Let X be any non-empty set. The **anti-discrete topology** on X is defined as $\mathcal{U} = \{\emptyset, X\}$. Only $\emptyset$ and X are open. **This example is not defined by a metric.**

Example 6.1.4 (Finite complement topology). Let X be an infinite set, say, $\mathbb{R}$. Define

$$U \in \mathcal{U} \iff X \setminus U \text{ is finite or } U = \emptyset. \quad (6.2)$$

Property 6.1.1. Let (X, d) be a metric space. Let $x, y \in X$, then there exists open sets U, V such that $x \in U$, $y \in V$ and $U \cap V = \emptyset$.

Definition 6.1.2. Let $(X, \mathcal{U})$ is a Hausdorff space (separable space) if for any $x, y \in X$, there exists open sets $U, V \in \mathcal{U}$ such that $x \in U$, $y \in V$ and $U \cap V = \emptyset$.

Remark Finite complement topology in 6.1.4 is not Hausdorff. If we take any two non-empty open sets U, V , then both $X \setminus U$ and $X \setminus V$ are finite. Thus, $X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$ is also finite, which means $U \cap V \neq \emptyset$. So it is not Hausdorff.

Remark Also we see that 6.1.3 is not Hausdorff.

Example 6.1.5. Let $X = \mathbb{N}$. Let $U_n = \{i \in X \mid i \leq n\}$ for some n . Let $\mathcal{U} = \{\emptyset, X\} \cup \{U_n \mid n \in \mathbb{N}\}$. Then $(X, \mathcal{U})$ is a topological space.

For finite Union, let $U_{n_1}, U_{n_2}, \dots, U_{n_k} \in \mathcal{U}$, then

$$\bigcup_{i=1}^k U_{n_i} = U_{\max\{n_1, n_2, \dots, n_k\}} \in \mathcal{U}. \quad (6.3)$$

For infinite Union, let $\{U_{n_\alpha}\}_{\alpha \in A} \subseteq \mathcal{U}$, then

$$\bigcup_{\alpha \in A} U_{n_\alpha} = \mathbb{N} \quad (6.4)$$

For finite Intersection, let $U_{n_1}, U_{n_2}, \dots, U_{n_k} \in \mathcal{U}$, then

$$\bigcap_{i=1}^k U_{n_i} = U_{\min\{n_1, n_2, \dots, n_k\}} \in \mathcal{U}. \quad (6.5)$$

This set is not Hausdorff.

Alternatively, topology can be defined by closed sets: $Y \subseteq X$ is closed if $X \setminus Y$ is open. So we can rewrite the definition of Topology space:

1. $\emptyset, X$ are closed.
2. If F_α is closed for all $\alpha \in A$, then $\bigcap_{\alpha \in A} F_\alpha$ is closed.
3. If $F_1, F_2, \dots, F_n$ are closed, then $\bigcup_{i=1}^n F_i$ is closed.

Question Is there an infinite collection of open sets such that their intersection is not open?

6.2 Zariski Topology

Let $X = \mathbb{C}^n$ (or $\mathbb{K}^n$). $Y \subseteq \mathbb{C}^n$ is closed if Y is a solution of

$$\begin{cases} f_1(x_1, x_2, \dots, x_n) = 0 \\ f_2(x_1, x_2, \dots, x_n) = 0 \\ \vdots \\ f_m(x_1, x_2, \dots, x_n) = 0 \\ \vdots \end{cases} \quad (6.6)$$

X is defined by $0 = 0$. $\emptyset$ is defined by $1 = 0$. Any points in $\mathbb{C}^n$ is defined by

$$\begin{cases} x_1 - a_1 = 0 \\ x_2 - a_2 = 0 \\ \vdots \\ x_n - a_n = 0 \end{cases} \quad (6.7)$$

A circle in $\mathbb{C}^2$ is defined by $x^2 + y^2 - 1 = 0$.

Any intersection of closed sets is the set of solutions of all the polynomials that define each closed set, so it is still closed.

Any union of finite closed sets is also closed since we can use the product of all polynomials that define each closed set to define the union.

For $n = 1$, since any polynomial with degree at least 1 has finitely many roots, recalling the finite complement topology in 6.1.4, we see that Zariski topology on $\mathbb{C}$ is the finite complement topology. But this is not Hausdorff.